Calculus II Lesson 25 Conics Sections - Parabola

Equations of Parabola:

1) Parabola opens up: $(x-h)^2 = 4p(y-k)$ ∨

vertex $=(h,k)$; focus $=(h,k+p)$; directrix: $y = k-p$

2) Parabola opens down: $(x-h)^2 = -4p(y-k)$ ∧

vertex $=(h,k)$; focus $=(h,k-p)$; directrix: $y = k+p$

3) Parabola opens right: $(y-k)^2 = 4p(x-h)$ <

vertex $=(h,k)$; focus $=(h+p,k)$; directrix: $x = h-p$

4) Parabola opens left: $(y-k)^2 = -4p(x-h)$ >

vertex $=(h,k)$; focus $=(h-p,k)$; directrix: $x = h+p$

1) Parabola: $y^2 = 10x$

Hint: Matching Formula: $(y-k)^2 = 4p(x-h)$

Write $y^2 = 10$ in standard form $\Rightarrow$ $(y-0)^2 = 10(x-0)$

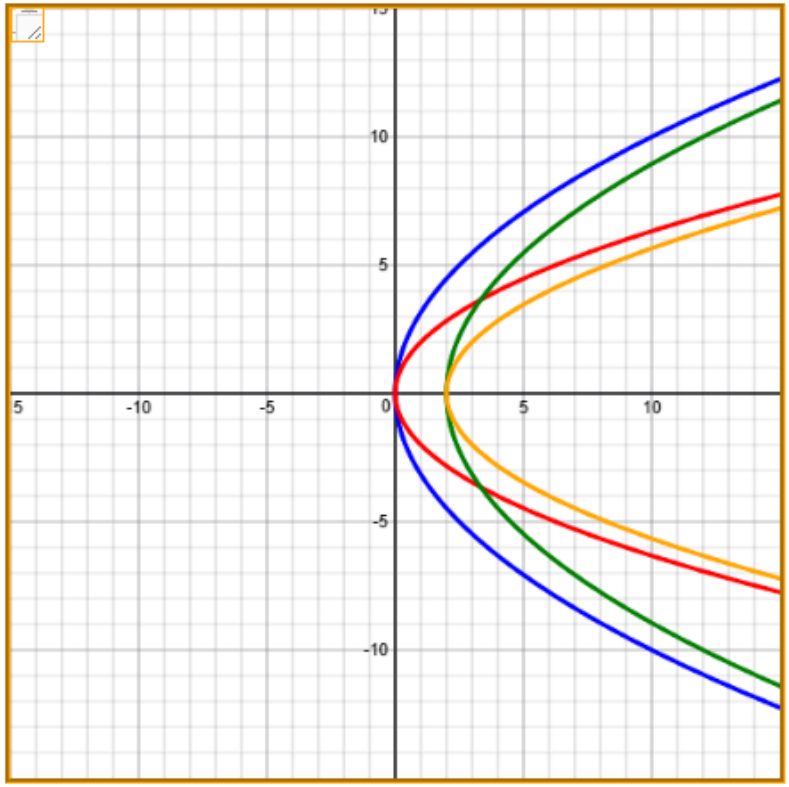
a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph $y^2 = 10x$ (multiple choice).

Choose the correct graph (blue, red, green, or orange)



2) Parabola: $y^2 - 6y = 10x$

Hint: Matching Formula: $(y-k)^2 = 4p(x-h)$

Write $y^2 - 6y = 10x$ in standard form:

Half of $-6 = -3$ and $(-3)^2 = 9$;

Add 9 to both sides of $y^2 - 6y = 10x$

$$\Rightarrow \quad y^2 - 6y + 9 = 10x + 9$$

$$\Rightarrow \quad (y-3)(y-3) = 10x + 9$$

$$\Rightarrow \quad (y-3)^2 = 10\left(x + \frac{9}{10}\right)$$

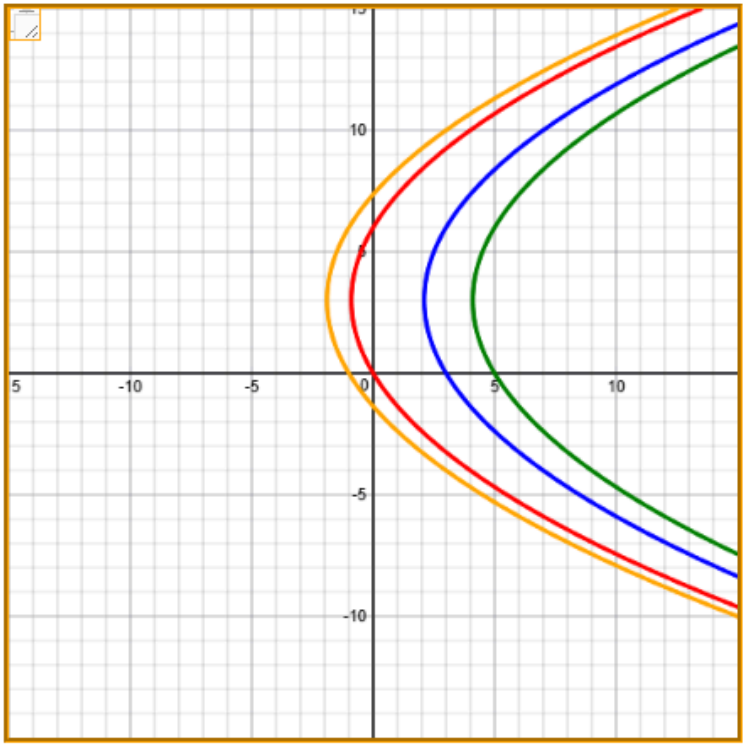
a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph $y^2 - 6y = 10x$ (multiple choice).

Choose the correct graph (blue, red, green, or orange)



3) Parabola: $x^2+8x+8y-10=0$

Hint: Matching Formula: $(x-h)^2=-4p(y-k)$

Write $x^2+8x+8y-10=0$ in standard form:

Move $8y-10$ to the right side: $x^2+8x=-8y+10;$

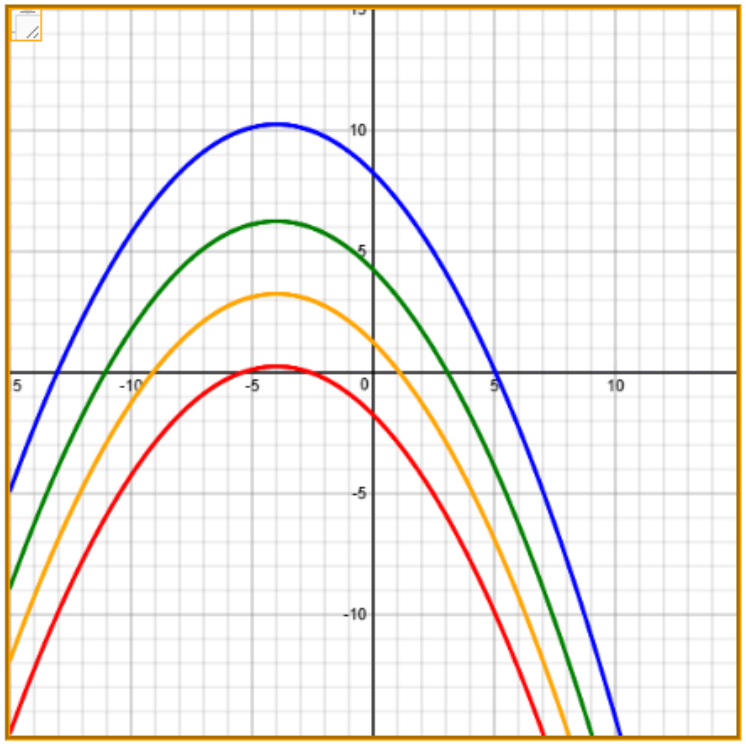
Half of $8=4$ and $(4)^2=16;$

Add 16 to both sides of $x^2+8x=-8y+10$

$$\Rightarrow \quad x^2+8x+16=-8y+10+16$$

$$\Rightarrow \quad (x+4)(x+4)=-8y+26$$

$$\Rightarrow \quad (x+4)^2=-8\left(y-\frac{26}{8}\right)$$



a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph $x^2+8x+8y-10=0$ (multiple choice).

Choose the correct graph (blue, red, green, or orange)

4) Find an equation of the parabola that has the following vertex and focus: vertex: (4, 3) ; focus: (1, 3)

Hint: Matching Formula: $(y-k)^2 = -4p(x-h)$

a) p = distance from vertex to directrix
= distance from vertex to focus = ____

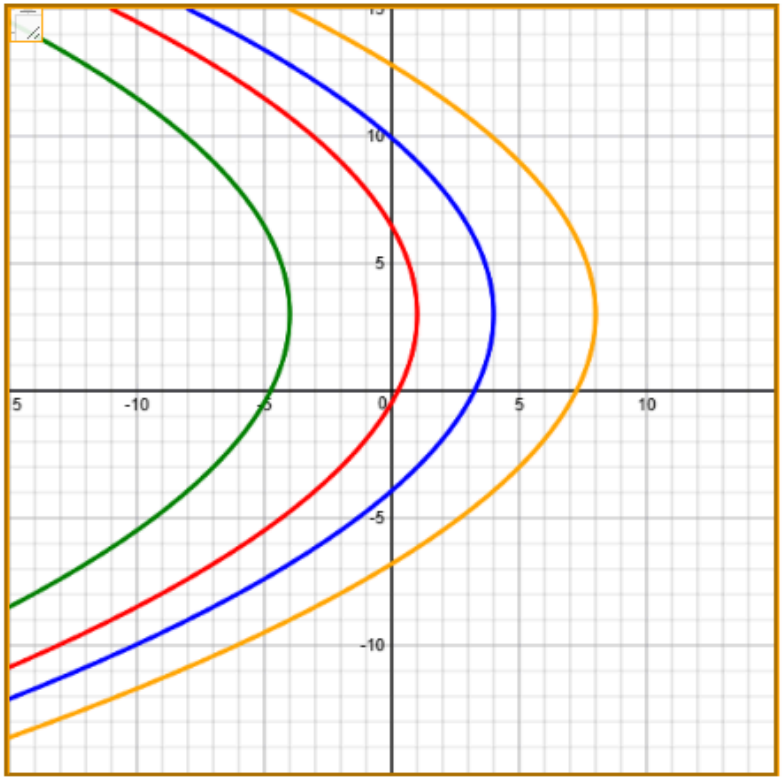
b) Equation of parabola:_____ (multiple choice)

A) $(y-3)^2 = -6(x-4)$ B) $(y-3)^2 = -8(x-4)$

C) $(y-3)^2 = -12(x-4)$ D) $(y-3)^2 = -14(x-4)$

c) Graph equation of parabola. (multiple choice)

Choose the correct graph (blue, red, green, or orange)



5) Find an equation of the parabola that has the following vertex and directrix: vertex: $(4, 3)$; directrix: $x = 2$

Hint: Matching Formula: $(y-k)^2 = 4p(x-h)$

a) p = distance from vertex to directrix = _____

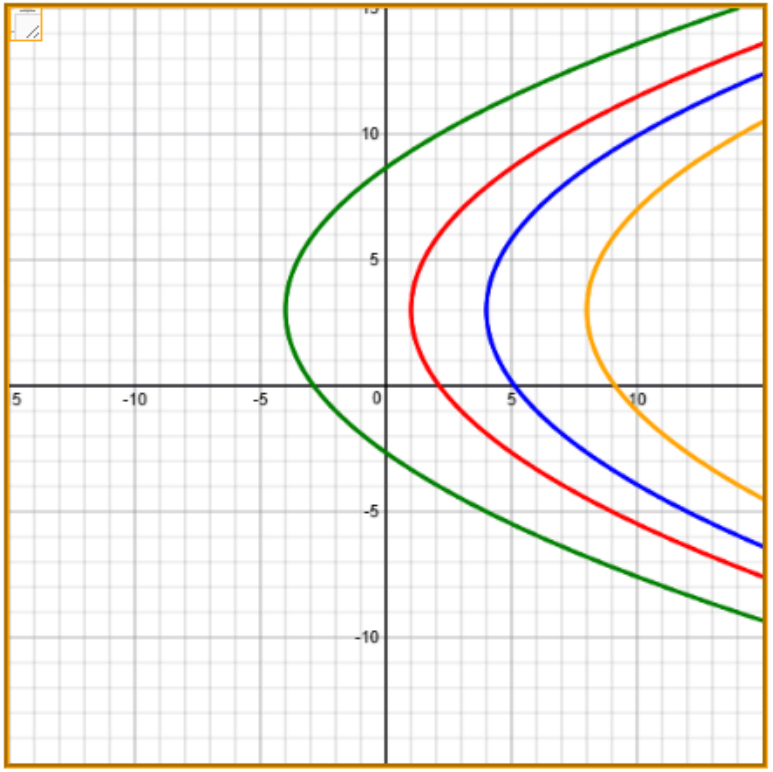
b) Equation of parabola:_____ (multiple choice)

A) $(y-3)^2 = 2(x-4)$ B) $(y-3)^2 = 3(x-4)$

C) $(y-3)^2 = 8(x-4)$ D) $(y-3)^2 = 5(x-4)$

c) Graph equation of parabola. (multiple choice).

Choose the correct graph (blue, red, green, or orange)



6) Parabola: $x^2 - 4x + 4y - 12 = 0$

Hint: Matching Formula: $(x-h)^2 = -4p(y-k)$

Write $x^2 - 4x + 4y - 12 = 0$ in standard form:

Move $4y - 12$ to the right side: $x^2 - 4x = -4y + 12$;

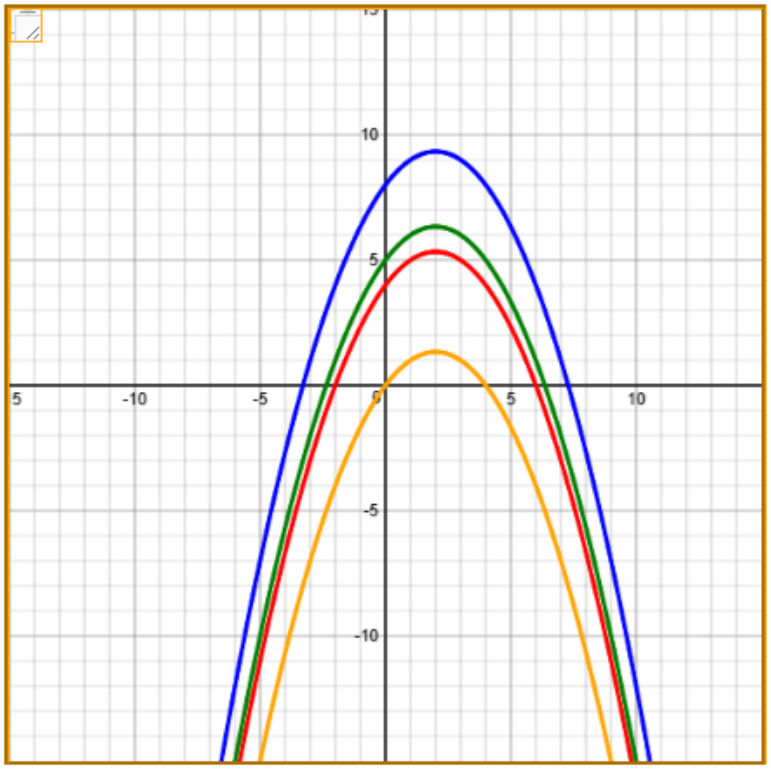
Half of $-4 = -2$ and $(-2)^2 = 4$;

Add 4 to both sides of $x^2 - 4x = -4y + 12$

$$\Rightarrow \quad x^2 - 4x + 4 = -4y + 12 + 4$$

$$\Rightarrow \quad (x-2)(x-2) = -4y + 16$$

$$\Rightarrow \quad (x-2)^2 = -4(y-4)$$



a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph equation of parabola. (multiple choice).

Choose the correct graph (blue, red, green, or orange)

7) Parabola: $y^2 - 4x + 2y - 8 = 0$

Hint: Matching Formula: $(y-k)^2 = 4p(x-h)$

Write $y^2 - 4x + 2y - 8 = 0$ in standard form:

Move $-4x - 8$ to the right side: $y^2 + 2y = 4x + 8$

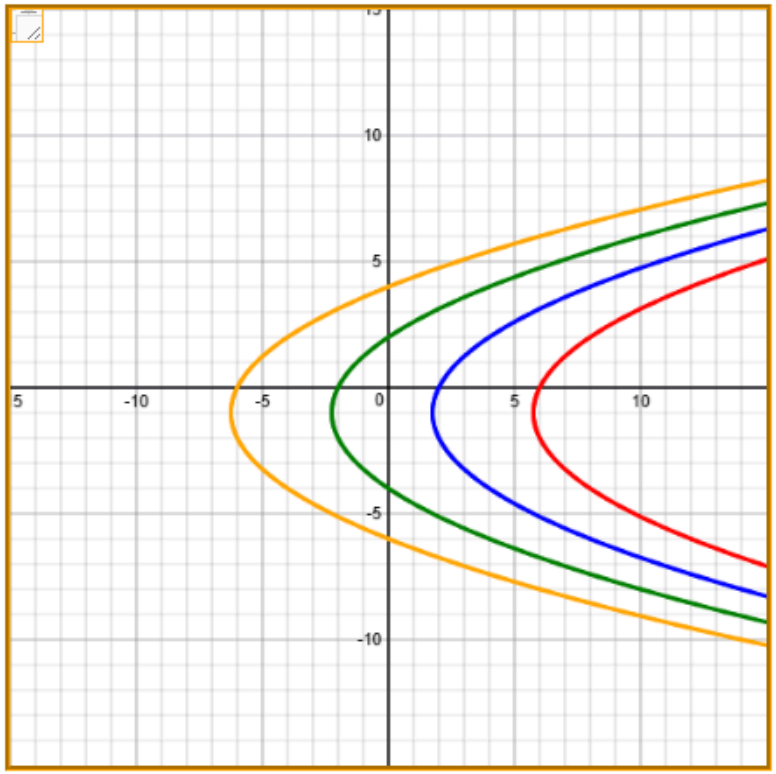
Half of $2 = 1$ and $(1)^2 = 1$;

Add 1 to both sides of $y^2 + 2y = 4x + 8$

$$\Rightarrow \quad y^2 + 2y + 1 = 4x + 8 + 1$$

$$\Rightarrow \quad (y+1)(y+1) = 4x + 9$$

$$\Rightarrow \quad (y+1)^2 = 4\left(x + \frac{9}{4}\right)$$



a) Find the vertex:_____

b) Find the focus:_____

c) Find the directrix:_____

d) Graph equation of parabola. (multiple choice).

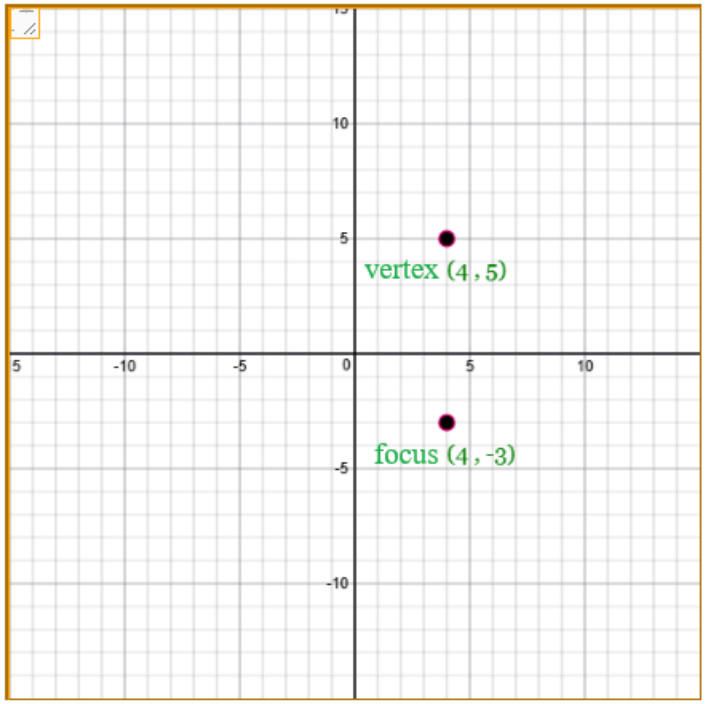
Choose the correct graph (blue, red, green, or orange)

8) Find an equation of the parabola that has the following vertex and focus:

vertex: (4, 5) ; focus: (4 , -3)

Equation of parabola:_____

Hint: Matching Formula: $(x-h)^2 = -4p(y-k)$

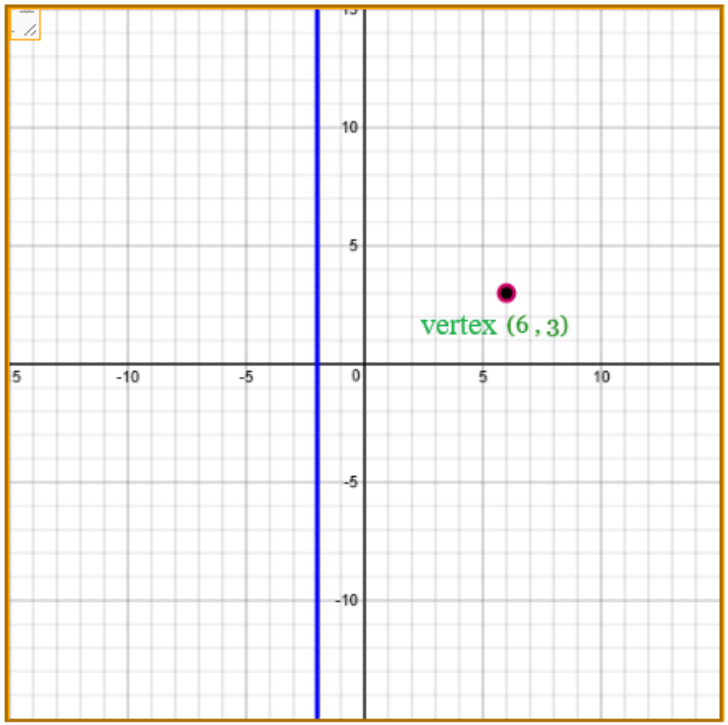


9) Find an equation of the parabola that has the following vertex and directrix:

vertex: $(6, 3)$; directrix: $y = -2$

Equation of parabola:_____

Hint: Matching Formula: $(y-k)^2 = 4p(x-h)$



10) A baseball was hit 3 feet from the ground level with speed of 120 feet/sec and angle of elevation is $\theta = 30^\circ = \pi/6$ rad.

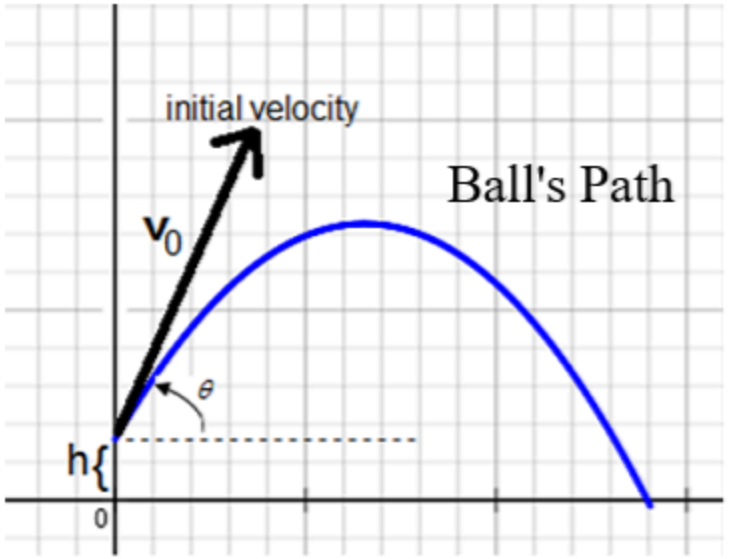
Equation for the ball's path:

$$y = h + v_0 \sin\theta\left[\frac{x}{v_0\cos\theta}\right] - \frac{1}{2}\cdot g\left[\frac{x}{v_0\cos\theta}\right]^2; \qquad g = \text{Earth's gravity} = 32 \text{ ft/sec}^2$$

$$y = 3 + 120\sin(\pi/6)\left[\frac{x}{120\cos(\pi/6)}\right] - \frac{1}{2}\cdot 32\left[\frac{x}{120\cos(\pi/6)}\right]^2$$

$$y = 3 + 60\left[\frac{x}{60\sqrt{3}}\right] - \frac{1}{2}\cdot 32\left[\frac{x}{60\sqrt{3}}\right]^2 = 3 + \frac{1}{\sqrt{3}}x - \frac{16}{\left(60\sqrt{3}\right)^2}x^2$$

$$y = 3 + 0.57735x - 0.00148148148x^2$$



Find the maximum height reached by the ball.

a) $y' =$ _____?_____

b) Set $y' = 0$ and solve for x. $x =$ _____?_____

c) Ball reached maximum height at the point: $x =$ _____?_____ and $y =$ _____?_____